Smart Temperature Monitoring System Using Arduino and DHT11

Abstract

The Internet of Things (IoT) has transformed the way devices interact and communicate. One of the most common applications of IoT is environmental monitoring using sensors. This project presents a Smart Temperature Monitoring System using an Arduino Uno and a DHT11 sensor. The system measures temperature and humidity from the surrounding environment and displays the readings on the Serial Monitor. The project demonstrates how sensor data can be collected and monitored in real time. Such systems are widely used in smart homes, weather stations, agriculture, healthcare, and industrial monitoring applications.

Introduction

Temperature and humidity monitoring are essential in many fields such as agriculture, healthcare, industrial automation, and environmental studies. Traditional monitoring methods require manual observation, which can be time-consuming and less accurate. With the advancement of IoT technologies, sensor-based monitoring systems have become more efficient and reliable.

The Smart Temperature Monitoring System uses a DHT11 sensor connected to an Arduino Uno microcontroller. The sensor continuously measures temperature and humidity and sends the data to the Arduino. The Arduino processes the information and displays the readings on the Serial Monitor. This project serves as a basic IoT prototype that demonstrates real-time environmental monitoring.

Objective

The main objectives of this project are:

- To measure temperature and humidity using a DHT11 sensor.
- To display sensor readings in real time.
- To understand the working of IoT-based monitoring systems.
- To develop a simple and low-cost environmental monitoring solution.

Components Used

Component	Quantity
Arduino Uno	1
DHT11 Sensor	1
Breadboard	1

Component	Quantity
Jumper Wires	As Required
USB Cable	1

Circuit diagram

Circuit Connections

The DHT11 sensor is connected to the Arduino Uno as follows:

DHT11 Pin	Arduino Pin
VCC	5V
GND	GND
DATA	Digital Pin 2

The Arduino supplies power to the sensor and receives temperature and humidity data through the data pin.

Working Principle

The DHT11 sensor continuously senses the temperature and humidity of the surrounding environment. The sensor converts the measured values into digital signals and sends them to the Arduino through the data pin.

The Arduino reads the sensor values using the DHT library and processes the information. The measured temperature and humidity values are then displayed on the Serial Monitor. The process repeats every two seconds, providing real-time environmental data.

This simple monitoring system demonstrates how IoT devices can collect and display sensor information for various practical applications.

Arduino Program

```
#include <DHT.h>

#define DHTPIN 2

#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {

  Serial.begin(9600);

  dht.begin();

}
```

```
void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
  Serial.print("Temperature: ");
  Serial.print(temp);
  Serial.print(" °C ");
  Serial.print("Humidity: ");
  Serial.print(hum);
  Serial.println(" %");
  delay(2000);
}
```

Output

The Serial Monitor displays the measured temperature and humidity values as shown below:

Temperature: 29°C Humidity: 65%

Temperature: 30°C Humidity: 64%

Temperature: 31°C Humidity: 63%

The readings are updated every two seconds, allowing continuous monitoring of environmental conditions.

Applications

- Smart Home Automation
- Weather Monitoring Stations
- Greenhouse Monitoring
- Industrial Environment Monitoring
- Healthcare Monitoring Systems
- Agricultural Applications

Advantages

- Low-cost implementation.
- Easy to build and operate.
- Real-time monitoring capability.

- Accurate measurement of environmental conditions.
- Can be integrated with IoT cloud platforms for remote monitoring.

Conclusion

The Smart Temperature Monitoring System successfully demonstrates the use of Arduino and DHT11 sensors for real-time environmental monitoring. The project provides accurate temperature and humidity measurements and displays them continuously on the Serial Monitor. This system can be further enhanced by integrating Wi-Fi modules, cloud platforms, and mobile applications for remote monitoring. The project serves as an excellent introduction to IoT-based sensor monitoring systems.

References

1. Arduino Official Documentation.
2. DHT11 Sensor Datasheet.
3. Arduino IDE User Guide.
4. Internet of Things: Principles and Applications.
5. ESP32 and Arduino Sensor Monitoring Tutorials.